

COINS FOR ANYTHING - CUSTOM COIN BUILDER

Is our AI coin photography state of the art?

A plain-English breakdown, with a side-by-side test on a real coin design - 17 September 2026

The short answer

Yes, as of today. The builder now renders with the newest image model your OpenAI account can reach (**gpt-image-2.5-sunburst**, released 4 September 2026). I checked the account directly: nothing newer exists on it. In a test on the same coin, the three newest-generation renders had every letter right and kept the logo intact; the older setup did not.

But the model is only about a third of what makes the photos impressive. The rest is what we feed it and how we check its work, and that is where most of the recent gains came from. This document covers all of it, what it costs, what I recommend changing next, and how to make sure you stay on the best technology as new models come out.

What "state of the art" means for a coin

Leaderboards rank image models on how pretty their pictures are. That is not your problem. A customer deciding whether to pay for 100 coins is judging four things, in this order:

What the customer judges	Why it matters	Where we stand
1. Their words, spelled right	One wrong letter and the whole proof looks amateur. This was your original complaint.	Letter-perfect in every new-generation test
2. Their logo, unchanged	A redrawn logo is a different logo. Brands notice immediately.	Kept exact, down to the (R) mark
3. It looks like real struck metal	Relief, plating, reflections. This is the "wow".	Strong; can be sharper (see Recommendation 2)
4. It arrives quickly	People leave if a preview takes minutes.	About 40 seconds; account limits need raising

THE EVIDENCE

Same coin, six ways

I gave six model and quality combinations the identical job: the same art proof, the same instructions, the same two reference photos of your real coins. The design is deliberately hard: a 40-character legend around the top, a second legend around the bottom, center wording wrapped over two lines, and a brand logo that must not change. Every result was then proofread automatically *and* checked by eye.



A. gpt-image-1, medium
What the app used before
Fails



B. gpt-image-2, high
Previous generation
Logo changed



C. gpt-image-2.5-flare, high
Newest, fast variant
Passes



D. gpt-image-2.5-sunburst, high
What the app uses now
Passes



E. gpt-image-2.5-sunburst, max
Newest, maximum quality
Passes



F. gpt-image-2.5-sunburst, 2048 px
Newest, double resolution
Passes, sharpest

What the test showed

	Model and setting	Time	Cost	Wording	Logo	Verdict
A	gpt-image-1, medium the old setup	22 s	n/a	Wrong: legend garbled at the end	Kept	Tilted, cropped at the edge, lettering printed flat black instead of struck metal. This is the look you complained about.
B	gpt-image-2, high	122 s	n/a	Correct	Changed: filled in white	Slowest of all, and it altered the brand mark. No reason to use it.
C	gpt-image-2.5-flare, high	27 s	8 cents	Correct	9 / 10	Fastest good result. A sensible fallback if speed ever matters more than finish.
D	gpt-image-2.5-sunburst, high current setting	37 s	8 cents	Correct	10 / 10	Clean, accurate, professional. The right default.
E	gpt-image-2.5-sunburst, max	104 s	24 cents	Correct	10 / 10	Slightly richer relief, but three times the cost and wait. Not worth it as the default.
F	gpt-image-2.5-sunburst, high, 2048 px	47 s	14 cents	Correct	10 / 10	Best value upgrade: visibly finer metal texture for 10 more seconds and 6 more cents.

Costs are my estimates from the token counts OpenAI returned for each render, at the published GPT Image 2.5 rates (\$30 per million image-output tokens, \$8 per million image-input tokens). Older models use a different price list, so I left them blank rather than guess. Proofreading adds a small amount per render on top. Please confirm current prices on OpenAI's pricing page before budgeting.

Standard versus double resolution, zoomed in



Left to right: D (standard), E (maximum quality), F (double resolution). F shows real brushed-metal grain and a finer surface; this is the difference a customer sees when they zoom in on their phone or download the image.

An honest note about the proofreader. During this test it made two mistakes. On render B it misread the date "1918" as "1978" on one of its two passes and raised a false alarm (I checked the image myself: the date is correct). And it scored B's logo 10 out of 10 even though the logo had been filled in white. Neither mistake let a bad *word* through, and a false alarm only costs one extra render. But it is a strict second pair of eyes, not a guarantee, which is why your team reviewing every design before production still matters.

HOW IT WORKS

The model is one part of seven

Switching to the newest model helped, but it was not the biggest fix. Image models are excellent at metal and unreliable at spelling, so the builder is designed around not trusting them. Each step below removed a specific cause of bad renders.

Step	What happens	The problem it solved
1. Clean references	Only photos of real finished coins are shown to the AI as style examples.	The folder had filled up with test logos and even a screenshot of a shipping form. The AI was being told those were "coins we made".
2. Smart layout	Lettering is measured with the real font and fitted to the circle: legends resize as a pair, long wording wraps, the logo keeps its true shape.	Long text ran off its path or into the inner ring, so the AI was handed a broken design to copy.
3. A crisp art proof	The AI receives a flat, high-contrast proof of the layout, not the soft embossed on-screen preview.	Soft, low-contrast lettering is exactly what image models misread.
4. A precise brief	The instructions are written on the server from the customer's real choices. Every line is quoted, counted and spelled letter by letter; the logo is declared off limits.	Vague instructions let the model improvise wording and "improve" logos.
5. Newest model, high quality	gpt-image-2.5-sunburst first, stepping down automatically only if OpenAI rejects a request.	The app was pinned to a model from April 2025 at medium quality: three generations behind.
6. Proofreading and retry	Each render is read back letter by letter, twice, and the logo is compared with the proof. A wrong render is redone with specific corrections.	Nobody was checking. Customers saw whatever came out.
7. Versions	Every render is kept with a tick or a warning. Picking an older one restores the design that made it.	A good render was lost the moment anything changed.

How we compare with the rest of the market

Other companies make strong image models. For this product they matter less than you might expect, because the builder does not ask an AI to *invent* a coin. It asks it to turn one specific approved layout into a photograph while copying style from your own coin photos. That needs a model that accepts several input images and edits faithfully, which narrows the field.

Model family	Known for	Fit for the coin builder
OpenAI GPT Image 2.5	Top of the blind-vote rankings for generation and editing; strong lettering; accepts multiple input images.	In use. Proven on your coins above.
Google Imagen 4 / Gemini image	Widely rated the most photorealistic.	The most credible challenger. Worth a bake-off if you ever want a second opinion on realism.
Ideogram v3	Reputation for the most reliable typography.	Interesting for text-only coins; less proven at keeping an uploaded logo exact.
FLUX.2 (Black Forest Labs)	Popular for logo and typography work; can be self-hosted.	An option if per-render cost ever becomes the priority.

This table is based on published roundups and rankings (September 2026), not on my own testing, so treat it as a map rather than a verdict. The only test that counts is the one on the previous pages: your coins, your logos, your wording. The project now includes a tool to repeat that test on any model in a few minutes.

WHAT TO DO NEXT

Recommendations, in priority order

1. Raise your OpenAI image limits before you launch **IMPORTANT**

Your account is currently limited to **5 input images per minute**. Each render sends three or four (the proof plus your reference photos), so the whole site can produce roughly one render a minute. My own test hit this limit. I have made the server wait and retry instead of showing an error, but with real traffic customers would queue. Limits rise automatically with account history and spend; you can see yours at platform.openai.com under Settings, Limits, and request an increase there.

2. Switch to double-resolution renders **RECOMMENDED**

One setting: **OPENAI_IMAGE_SIZE=2048x2048** in the .env file. It passed proofreading, added about 10 seconds and about 6 cents, and is the most visible quality gain available today. Test the full order flow once after switching, because the image files are about four times larger.

3. Invest an afternoon in reference photos **RECOMMENDED**

The AI copies the look of metal from the photos in the references folder, so their quality sets the ceiling. Best results come from straight-on, evenly lit photos of one coin each, sorted into folders by finish (antique-gold, shiny-silver and so on) so a gold order is styled from gold coins. One current photo shows two coins side by side and should be replaced. This costs nothing and improves every render.

4. Keep "max" quality for special moments, not as the default OPTIONAL

It is three times the price and takes well over a minute, for a subtle gain. A good use would be a single "final hero image" generated after the customer commits to ordering, for their receipt or for your marketing.

5. Ideas that would genuinely impress FUTURE

An angled "beauty shot" with depth and shadow alongside the straight-on proof; a render of the back of the coin; the coin shown in a hand or a presentation box for scale. All are achievable with the current model and pipeline.

STAYING AHEAD

How to make sure you stay state of the art

New image models now arrive every few months, and each generation has been a real step forward for lettering. The risk is not choosing the wrong model today; it is quietly falling behind, which is exactly what had happened before. Three safeguards are now built in:

- **The app reaches for the newest model by itself.** It tries the best model first and steps down only if OpenAI refuses a request, so it never gets stuck on an old one. When a new generation is released, adding it is a one-line change.
- **A free monthly check.** Running `npm run check-models` lists every image model your account can use and says plainly whether something newer than the one in use has appeared. It generates nothing and costs nothing. Today it reports: "Nothing newer is available."
- **A repeatable bake-off.** Running `npm run bakeoff` sends one art proof to any set of models, proofreads each result and produces a side-by-side page, the same comparison as in this document. It costs a few renders. Switch models only when a newcomer wins on *your* coins.

A simple routine: on the first of each month, run the free check. If it names a new model, run a bake-off on two or three real customer designs (one long legend, one detailed logo, one colorful logo). If the newcomer wins, switch the default. Ten minutes a month keeps you current.

Settings at a glance

Setting (.env file)	Now	What it controls
OPENAI_IMAGE_MODEL	not set = newest	Leave unset so the app always prefers the newest model.
OPENAI_IMAGE_QUALITY	high	Lettering accuracy drops below high. "max" is the hero-shot option.
OPENAI_IMAGE_SIZE	1024x1024	Recommendation 2: set to 2048x2048.
REFERENCE_COUNT	4 in your file (the app caps it at 3)	Style photos sent with each render. I suggest 2: more photos means more unrelated lettering for the AI to confuse, and more of your per-minute limit used.
AI_MAX_ATTEMPTS	2	A render that fails proofreading is redone once with corrections.
DEBUG_PROOF_DIR	off	Saves the exact art proof the AI received, for investigating a poor render.

Sources and method. Model availability and release dates were read directly from your OpenAI account on 17 September 2026. The six comparison renders, their timings and token counts are from live tests run that day; the images in this document are those renders, unretouched. Pricing rates and the market comparison come from public articles and rankings (llm-stats.com image leaderboard; MindStudio, Atlas Cloud and aimlapi.com roundups) and were not independently verified.